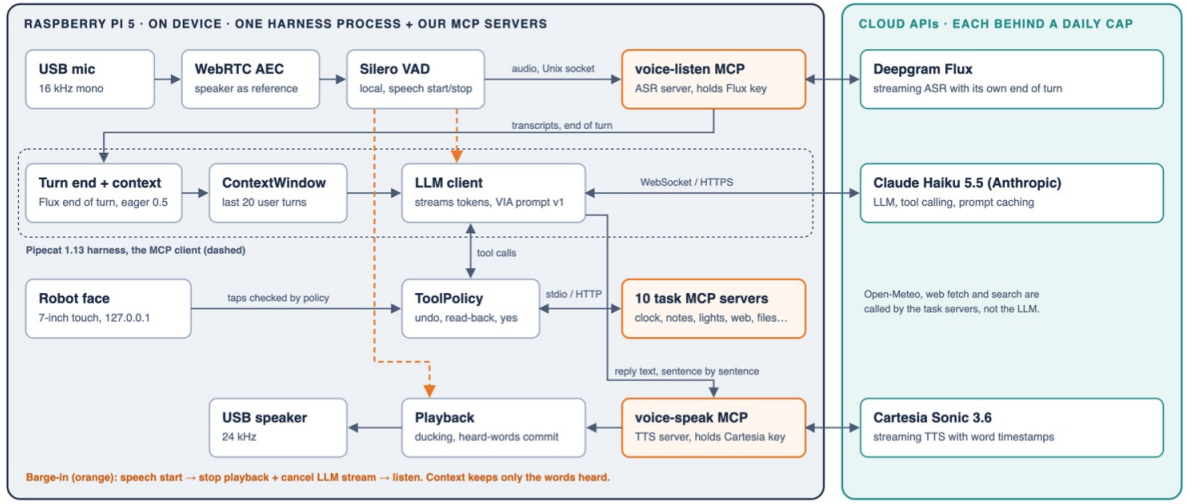
Full-Duplex Voice Agent on a Raspberry Pi

VIA ("voice input agent") · Nov 6, 2026

1 SYSTEM DIAGRAM the one diagram that explains the build




```
voice in → VAD → voice-listen → LLM (+MCP tools) → voice-speak → voice out

barge-in: speech detected → stop playback + cancel LLM stream → listen
```

- USB mic → WebRTC AEC → Silero VAD on the Pi → voice-listen MCP (ASR) → Pipecat agent → voice-speak MCP (TTS) → USB speaker
- Harness with MCP client: Pipecat 1.13, one process on the Pi
- LLM is a commercial API (Claude Haiku 5.5); every tool comes from our own MCP servers
- All 12 MCP servers are ours and run on the Pi: voice-listen, voice-speak and 10 task servers
- Barge-in stops playback and cancels the LLM stream; Cartesia word timestamps keep only the words heard
- On-Pi left, cloud right: Deepgram Flux, Anthropic, Cartesia, each behind a hard daily cap

2 PROMPT in the repo, versioned: harness/bot.py PROMPT_VERSION

SYSTEM PROMPT (VIA v1) - excerpt

```
You are VIA, short for voice input agent, a hands-free voice assistant, and every reply is spoken
aloud. Start with a short sentence of a few words, and keep the whole reply under about 30 words: no
lists, formatting or parentheses. Call a tool straight away, with no words before it. If a tool
result says waiting_for_undo or needs_confirmation, it has not happened yet: say so briefly and do
not call the tool again. Speech recognition may mishear: act on the likely meaning, but if a short
fragment could mean different things, ask in a few words and do not act.
```

Stopping on barge-in and confirming risky actions are enforced by the harness (ToolPolicy), not left to the model.

3 CONTEXT ENGINEERING

GOES IN:

- System prompt (VIA v1) + agent persona
- Flux final transcripts (partials only drive the face)
- Flux end-of-turn events, eager at 0.5
- MCP tool results + notes of screen taps
- Document agents: docs read tools (no vector store)

MANAGED:

- Last 20 user turns kept: 26 trims in 41 turns
- Cached stable prefix: ~87% less LLM cost
- idle → listen → think → speak, on the face
- Barge-in: queued audio dropped, heard words kept
- Logged: prompt tokens median 6.8K, max 8.5K

MCP Services, Evaluation & Model Serving

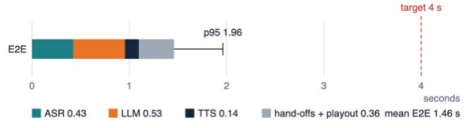
all servers built by us, running on the Pi

4 MCP SERVICES + section 6: model serving (one table, one truth)

Component	MCP server / transport	Tools	Model	Served where / notes
ASR	voice-listen / stdio, HTTP	listen, stop_listening, status	Deepgram flux-general-en	Pi; holds the key; Flux ends the turn
TTS	voice-speak / stdio, HTTP	speak, stop, status	Cartesia sonic-3.6	Pi; word timestamps; cut on barge-in
LLM	harness MCP client (Pipecat 1.13)	function calling over all task tools	claude-haiku-5-5	Anthropic API; streaming, prompt caching
VAD	in-process	speech start / stop	Silero VAD	Pi on-device, 16 kHz mono
Task	clock, notes, lights, thermostat	set_timer, add_note, delete_note, set_light, set_thermostat	none	Pi; lights, thermostat are mocks
Task	music, comms, delegate	play_music, next_track, send_message, call_contact, start_task	Spotify; Haiku 5.5 worker	Pi; comms is a mock by default
Task	knowledge, docs, workspace	get_weather, fetch_content, search, read_file, write_file, bash	Open-Meteo, web	Pi; workspace only; bash sandboxed

5 EVALUATION & BENCHMARK fd-benchmark stage test, the class's standard method

- One clock: the Pi's monotonic clock; fdbench's Pipecat observer stamps every stage
- VAD end → ASR final → LLM 1st token → TTS 1st audio → speaker
- E2E = speaker − VAD end; barge-in = stop − speech start
- 10 phrases × 2 runs (20 turns); mean + p95; JSON log per run



latency_chart.svg, fdbench run 2026-11-06_21-22-21

Metric	Definition	Target	Mean	p95
E2E latency	end of speech → audio out	< 4.0 s	**1.46 s**	**1.96 s**
ASR final	end of speech → transcript	< 0.8 s	**0.43 s**	**1.18 s**
LLM TTFT	transcript → first token	< 1.5 s	**0.53 s**	**0.70 s**
TTS TTFB	text → first audio chunk	< 0.6 s	**0.14 s**	**0.18 s**
Barge-in	speech → playback stopped	< 0.8 s	**0.70 s**	**0.79 s (n=2)**
WER	ASR accuracy on test set	< 15%	**5.5%**	-

DEMO (2.5 min, docs/live-demo.md)

- Real tasks, no push-to-talk: kitchen light, a 2-minute timer
- Barge-in: "wait, stop" cuts a story mid-sentence
- Slow API: weather call covered by the waiting phrase
- Screen tap + voice share one policy; a delete held, then cancelled

RESULTS

- Raspberry Pi 5 Model B, 8 GB; active cooler, fan; peak 62.8 C, never throttled
- Wi-Fi to the cloud (IPv6 off); Nov 6, 2026
- Pipecat 1.13.0, Flux flux-general-en, Haiku 5.5, Cartesia sonic-3.6, Silero, prompt VIA v1
- Every target met except ASR final p95

CHALLENGES (problem, cause, fix, proof) & REFERENCES

Problem	Cause	Fix	Proof
Slow replies: 2.35 s	EOT wait 0.85 s, LLM 1.0 s, long 1st line	Eager EOT; short first line	**1.46 s: EOT 0.43, LLM 0.53, TTS 0.14**
Choppy speech	Aura-2 at 0.43x real time	Cartesia TTS (ADR-015)	**Median gap 354 → 45 ms**
Bot cut itself off	Heard its own voice; no AEC in Pipecat	AEC + echo filter (ADR-042)	**27 speaker turns, no self-cut**
Turns stalled, up to 7.1 s	Smart Turn can't end a Flux turn	Flux owns end of turn (ADR-005)	**5/5 turns, 1 cut-in in 10 pauses**
Prompt grew every turn	Whole history sent every call	Keep last 20 turns (ADR-047)	**26 trims in a 41-turn run**
Runs lost to rate limits	Groq: 8K tokens/min, ~5K a call	Haiku 5.5 as LLM (ADR-025)	**95.3% passed, none lost**

modelcontextprotocol.io · docs.pipecat.ai · github.com/trueshotjames/ai231-me3-full-duplex-benchmarking · run logs in logs/